

Transforming FMCG Marketing with AI-Powered Predictive Analytics: Insights from Multi-Model Evaluation

Peuvailu Kamal

2nd Year - M. S Data Science

EdTech Division_Exafluence INC

Sri Venkateswara University
Tirupati, India

peruvailukamal@gmail.com

Yaram Lakshmi Nisoka

2nd Year - M. S Data Science

EdTech Division_Exafluence INC

Sri Venkateswara University
Tirupati, India

yarnnisoka@gmail.com

Kamasani Vishnuvardhan Reddy

2nd Year - M. S Data Science

EdTech Division_Exafluence INC

Sri Venkateswara University
Tirupati, India

vishnureddyk3333@gmail.com

Bharghav Madhiraju

M.S Data Science

EdTech Division_Exafluence INC

Sri Venkateswara University

Tirupati, India

bhargavmadhiraju123@gmail.com

Abstract— In the Fast-Moving Consumer Goods (FMCG) industry which is characterized by high level of competition, fluctuating consumer demand, and very thin profit margins success is a result of being able to predict market trends instead of just reacting to them. This case study looks at the key role of Predictive Analytics (PA) which is supported by Artificial Intelligence (AI) and Machine Learning (ML) in terms of it being a trigger for marketing innovation and strategic decision making within the FMCG sector. We present a practical example of a multi algorithmic AI system we developed for sales volume prediction which is a very important element of marketing return on investment, supply chain performance and total profit. Our approach is to systematize the training and evaluation of a set of regression models which include Linear Regression, Ridge Regression, Support Vector Regression (SVR), Random Forest, and Gradient Boosting on a large scale FMCG data which we include temporal and promotional variables in. We partition the data into train and test sets which in turn we use for in depth validation of the predictive models. We use key performance indicators like Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2) to very closely compare model performance. What we find from our study is that ensemble methods in particular Gradient Boosting and Random Forest do very well in outperforming traditional linear and kernel-based models at what they do which is to put forward complex non-linear associations between many marketing drivers and sales results. With advanced predictive tools, marketing is changing from guesswork to smart strategies based on data. This shift allows companies to engage with customers more personally by targeting them more accurately. It also helps businesses improve their trade promotions by predicting sales boosts and guiding decisions about pricing and launching new products. Additionally, using these predictions within inventory management systems can cut down on waste throughout the supply chain. This study shows that using strong AI-driven predictive analytics isn't just a nice-to-have anymore; it's essential for fast-moving consumer goods (FMCG) companies to stay competitive, adapt to market changes, and grow sustainably in today's digital world. The case study offered is a helpful roadmap for other businesses looking to implement similar AI solutions effectively.

Keywords: Predictive Analytics, Artificial Intelligence, Machine Learning, FMCG, Demand Forecasting, Marketing Innovation, Ensemble Methods, Support Vector Regression, Gradient Boosting, Random Forest.